

Explanation of Revisions

Resubmission of ARR submission 1656:

“Correction and Corruption Signals: A Controlled Study of Capability-Asymmetric Multi-Agent LLM Debate Under Adversarial Wrong-Peer Exposure”

Summary

We thank the three reviewers and the Area Chair. The meta-review’s decision text identified the crux precisely:

“Although the authors stated in their response that additional model runs have been completed, in-cycle ACL policies reserve these for future revisions, leaving the current paper scope under-supported.”

Every experiment the reviewers asked for had been run but could not be reported in an author response. This revision puts all of them *in the manuscript*, on the identical question pool, random seed, prompts, and judge cascade as the reviewed version, and adds three further experiments that go beyond what was requested.

What is new in the manuscript.

1. **A solo re-answering control (C1R)** — the causal baseline sTG4 and the AC identified as the central methodological gap. Section 4.2.
2. **An honest-peer control (C3H, C4H)** — the same weak models answering naturally, addressing Tf6M’s and sTG4’s objection that persona-anchored peers are adversarial distractors rather than natural low-capability output. Section 4.2.
3. **A confidence-filter re-run with confidence actually elicited (C5R, C5H)** plus a corrected pre-flight test — the “properly designed follow-up” Tf6M asked for. Section 3.4, Section 4.4.
4. **An eight-model capability sweep** — the systematic sweep including open-weight models that Tf6M argued was necessary to show a general phenomenon rather than a pairwise anomaly. Section 4.3, Table 5, Figure 1.
5. **GPT-4o-mini extended from 50 to the full 300-question pool**, answering gL73’s concern about limited questions.
6. **A contamination probe** on numerically perturbed GSM8K.
7. **A controlled split-peer condition (C4split)** — tests whether a visible correct peer changes harmful revision, removing the last “deferred to future work” in the previous version. Section 4.6.
8. **A heterogeneous-smart debate control (C2het)** — three architecturally distinct strong models, converting the “homogeneous debate is close to sampling” limitation into a result.
9. **An output-distribution probe** on an open-weight focal model, measuring the probability mass that moves onto the peer-asserted wrong answer between rounds — evidence beneath the discrete choice. We call it an output-distribution probe, not a mechanism: it observes output probabilities and does not identify a representation-level cause.

What is new in the writing. The term *focal model* is now defined at first use in both the abstract and Section 1 (gL73 W1); Section 1 is restructured around three explicit research questions and the contributions each end with a pointer to the section, table or figure that supports it (gL73 W2, W5); the conditions are given in a table rather than in prose; and Section 2.2 gains a paragraph on **debate as an error detector**, the specific prior-work area the meta-review named as under-discussed (M7, gL73 W4).

Corrections we made to the previous version. In preparing this revision we re-derived every number in the manuscript directly from the released per-trial logs, and corrected fourteen discrepancies that survived the reviewed version. The full list is in the released `audit/numerical_consistency_report.md`; the ones that change a reported number are:

- Table 3 previously reported *raw* McNemar p -values under a caption stating they were Holm-corrected. Both are now reported in separate columns. All conclusions that were claimed as significant remain significant after correction.
- The contamination-probe contrast was reported as $p = 0.07$ without noting that this question-level test rests on eight discordant pairs. Section 4.5 now states plainly that the probe is underpowered at the question level ($p = 0.23$ after correction) while significant at the trial level, and no longer claims significance for it.
- Appendix S1’s trial-count table still carried the superseded 50-question GPT-4o-mini numbers, which contradicted Table 2. All data tables are now generated programmatically from the released logs by a released script, so they cannot drift from the data again.
- Appendix S7 asserted that the weak models’ unprompted accuracy was “not measured in this study”. The honest-peer conditions measure it directly; the numbers are now given.
- The released corrected-gate report substituted zero for $P(\text{loud} \mid \text{correct})$ on the wrong-anchored substrate, where that class is empty by construction. This produced an apparent discriminative gap of 0.99 and a spurious “passed” verdict in the artefact, contradicting the paper’s own table. The report now records the substrate as undefined and gives per-class counts.
- The capability-sweep Spearman ρ was computed on the 1-dp rounded table values, which tie four mid-range models together and inflated it to -0.97 with $p = 0.0001$. On unrounded rates it is $\rho = -0.95$ with exact permutation $p = 0.0011$.
- Persona validation counts in Appendix S5 were each off by 14 (the failures were double-counted): the correct split is 1,116 passing first, 370 after regeneration, 14 failing.
- The primary focal model was described as being of the DeepSeek-V3 family with a citation to that technical report. The API resolves the requested alias to the served snapshot `deepseek-v4-flash`, for which no such report exists, so the manuscript now reports the served snapshot recorded in the run metadata and makes no architectural claim.

Point-by-point response

Meta-review (Area Chair QTRJ)

Point	Status	Where it is answered
Only two focal models; a systematic sweep across a wider set, including open-weight models, is necessary to show a general phenomenon.	New experiment	Eight focal models spanning 44.9–76.2% solo accuracy, on the same 300-question pool. All eight point estimates of the accuracy change under two confidently wrong peers are non-negative, and harmful revision rises steeply as solo accuracy falls (Spearman $\rho = -0.95$, exact permutation $p = 0.001$; leave-one-model-out $\rho \in [-0.96, -0.93]$; $\rho = -0.94$ excluding the two weakest). Section 4.3, Table 5, Appendix Figure.
Persona prompts turn weak peers into deliberate adversarial distractors; the manuscript remained scoped to distractors.	New experiment Scoped	Honest-peer conditions (C3H, C4H) run the same weak models with the standard prompt. Their unprompted accuracy is 50.4% and 45.1%. Honest peers leave focal accuracy at solo level (-0.3 pp, $p = 1.00$) and are clearly below the wrong-anchored condition (-7.3 pp, $p = 0.004$ Holm). The claims are scoped to confidently wrong peers throughout. Section 3.1, Section 4.2.
Major flaw in causal attribution: the $+7.9$ pp gain cannot be attributed to peer exposure without a re-examine/re-answer control.	New experiment	C1R gives the focal model the Round-1 revision instruction with no peer messages. Re-answering alone is indistinguishable from solo ($+0.3$ pp, $p = 1.00$); adding two wrong peers on top of that same second attempt gains $+6.7$ pp ($p = 0.011$ Holm). Section 4.2, Table 3.
C5 removed 100% of peer messages because the weak-peer prompt omitted the Round-0 confidence instruction; this renders C5 uninformative.	New experiment	The persona prompt was repaired and the filter re-run on peers that do state confidence (C5R), plus an honest variant (C5H). Peers now emit confidence in essentially every Round-0 message. The filter still removes almost nothing, now for a <i>signal</i> reason rather than a parsing one: confidence does not separate right from wrong (AUROC 0.58). Section 3.4, Section 4.4, Appendix S3, Appendix S4.
“Focal agent” left undefined in the main text.	Addressed	Defined at first use in the abstract and in Section 1.
Presentation of the research questions was unclear.	Addressed	Explicit RQ1–RQ3 in Section 1, each mapped to the section that answers it.

Point	Status	Where it is answered
Prior work on collaborative debate in error detection insufficiently discussed.	Addressed	New paragraph in Section 2.2 covering cross-examination as a factual-error detector, divergence-preserving debate, and the self-correction result that motivates external signal, with an explicit contrast: those works treat the peer as an examiner whose disagreement flags an error, whereas we find a <i>confidently wrong</i> peer supplies a usable prompt to recompute.

Reviewer gL73

Weakness	Status	Response
W1. “What is the ‘focal agent’?”	Addressed	The focal model is the single designated agent whose Round-0 and Round-1 answers are scored; all other agents are peers whose Round-0 messages it sees. Defined in the abstract and at first use in Section 1.
W2. The writing is poor and the main research problems are unclear.	Addressed	Section 1 rebuilt around RQ1–RQ3; the five-condition setting moved from prose into Table 1 with a plain-language “what it answers” column; statistics moved out of prose into tables; the abstract rewritten without condition codes.
W3. Experiments limited to a limited number of questions and LLMs.	New experiment	Eight focal models instead of two, and GPT-4o-mini extended from 50 to all 300 questions (1,500 trials per condition instead of 250). The headline harmful-flip rise for GPT-4o-mini is now a well-powered 7.4% → 13.4%.
W4. A closely related work has not been sufficiently discussed.	Addressed	We asked which work was meant; the meta-review identified the area as collaborative debate for error detection. Section 2.2 now discusses it (see M7 above). Section 2.2 also already added adversarial and capability-asymmetric debate.
W5. Unclear how the experiments support the listed contributions.	Addressed	Three contributions instead of four, each ending with the section, table or figure that supports it.
Datasets = 1, Software = 1, Reproducibility = 2.	Addressed	The full pipeline, all prompts, the question and persona pools, the per-trial logs for every trial, and the scripts that regenerate every number and figure are released now rather than on acceptance, through an anonymised repository linked in the Reproducibility section. The data tables in the paper are generated from those logs by a released script.

Reviewer Tf6M

Point	Status	Response
W1. The main claim rests on only two focal models; a wider sweep including open-weight models is needed to establish a general phenomenon rather than a pairwise difference. The paper does not explore whether the difference is predictable from a capability metric.	New experiment	The sweep spans eight focal models, six of them open-weight. The qualitative difference <i>is</i> predictable from a capability metric: solo accuracy predicts the harmful-flip rate almost monotonically (Spearman $\rho = -0.95$, exact permutation $p = 0.001$, $n = 8$), from 6.5% for the strongest model to 24.9% for the weakest. Notably the <i>gain</i> does not track ability ($\rho = 0.33$, $p = 0.42$) — it is the cost, not the benefit, that is graded by strength, which is a sharper claim than the reviewed version made. Section 4.3, Table 5.
W2. Anchored peers should be read as adversarial distractors, not honest low-capability outputs.	Scoped New experiment	We agree, and now demonstrate the distinction rather than only stating it. The honest-peer control shows the correction effect is specific to confidently wrong peers. Every conclusion is scoped accordingly, in the abstract, Section 1, Section 4.2 and Limitations.
W3. The confidence-weighting experiment is uninformative because the persona prompt omits the confidence line, giving 100% peer exclusion; the failure cannot be attributed to the filtering mechanism.	New experiment	Correct, and now fixed. With confidence elicited, the filter is tested on peers that state it, and still fails — because weak peers are equally loud when right and when wrong. We also give a corrected pre-flight test on the discriminative gap $P(\text{loud} \mid \text{wrong}) - P(\text{loud} \mid \text{correct})$, which fails in advance and would have predicted the null, whereas the original loud-when-wrong test passes at 0.995 and does not.
Suggestions 1–4 (re-answering control; honest weak peers; re-run the filter with reliable Round-0 confidence; expand to more focal models).	New experiment	All four carried out and reported: C1R; C3H/C4H; C5R/C5H with the corrected gate; the eight-model sweep.

Reviewer sTG4

Point	Status	Response
W1. The study is not about natural capability asymmetry — the weak peers are deliberately wrong distractors, not natural mistakes.	Scoped New experiment	The honest-peer conditions supply the natural-mistake comparison directly, and the claims are re-scoped. The honest condition also yields a composition gradient that the anchored design cannot produce: harmful flips run 25.9% when both peers happen to be wrong, 7.9% when they disagree, and 1.4% when both are right.

Point	Status	Response
W2 (main weakness). The +7.9 pp gain cannot be attributed to peer exposure without a control where the focal model simply re-examines the question.	New experiment	This is exactly C1R, and it is now the paper’s second experiment. Re-answering alone reproduces none of the gain. We had noted in our response that C5’s total peer removal made an accidental partial control; that argument is withdrawn in favour of the dedicated, powered experiment the reviewer asked for.
Suggestion 1: solo re-answering control with the same Round-1 instruction and reasoning budget but no peer messages.	New experiment	Implemented exactly as specified: identical Round-1 instruction and token budget, peer block absent. Section 3.1, Section 4.2.
Suggestion 2: evaluate honest weak peers not anchored to wrong answers.	New experiment	C3H and C4H, 3,000 trials.
Suggestion 3: rerun confidence filtering with a prompt that reliably elicits Round-0 confidence from all peers.	New experiment	C5R and C5H.
Suggestion 4: expand the analysis to more focal models with different capability.	New experiment	Eight focal models.

Beyond what was requested

Three experiments in this revision were not asked for. They close the two weaknesses the previous version conceded in its own Limitations, and add mechanistic evidence.

- **Split-peer control (C4split), 1,500 trials.** The reviewed version admitted its design could not separate “a wrong answer is present” from “every visible peer is wrong”, and deferred a split-peer control to future work. It is now run: one wrong-anchored and one correct-anchored peer, same pool and replications. Adding a correct peer lowers harmful revision (4.8% $\rightarrow$ 3.4% paired, $n = 1,042$, $p = 0.03$) and raises accuracy to 86.3% — +4.7 points over two wrong peers ($p = 0.013$ corrected) and +11.7 over solo — while the correction rate is unchanged (56.0% vs 55.8%). We initially read this as evidence that *unanimity* drives the harm, then tested that directly and found it false: within C4 the two anchored peers name the same wrong answer in only 16.4% of trials, and whether they agree makes no difference to harmful revision (5.95% vs 6.58%, $p = 0.88$). The manuscript therefore claims only that a *visible correct peer* reduces harmful revision, and states that C4split changes two things at once (removes a wrong peer, adds a correct one) so it isolates neither.
- **Heterogeneous-smart debate (C2het), 900 trials.** The reviewed version conceded that its homogeneous condition was closer to self-consistency sampling than to debate (pairwise $\kappa = 0.72$, ESS ≈ 1.2). Three architecturally distinct strong models now debate on the same pool. This heterogeneous control did not improve performance: accuracy is indistinguishable from solo (+0.7, $p = 0.87$), the focal ends below where it started (79.0% $\rightarrow$ 77.3%) where self-sampling gains (75.8% $\rightarrow$ 78.7%), and harmful

revision rises from 3.3% to 5.8% ($n = 638$, $p = 0.02$). Because both distinct peers are also 9–10 points weaker than the focal, diversity and peer strength are not separated and the control does *not* estimate the effect of diversity alone; the manuscript says so explicitly and keeps the corresponding limitation.

- **Mechanistic log-probability probe.** Limitations previously said the study was “behavioural, not mechanistic”. On an open-weight focal model served locally (*Llama-3.1-8B-Instruct*, 2,700 trials) we measure the probability mass that moves onto the peer-asserted wrong answer between rounds. Homogeneous peers move the model *away* from that answer (-0.018); two confidently wrong peers move it *toward* it ($+0.064$). Paired by (question, replication) the shift is $+0.082$ (95% CI $[+0.067, +0.097]$, Wilcoxon $p \approx 10^{-42}$, $n = 898$). The pull is therefore present and measurable on trials where the stated answer never changes — which flip rates structurally cannot show.

One finding that cuts against our own design. The honest-peer control turned out to do more than establish scope. Comparing trials where both weak peers were wrong, paired by question so difficulty is held fixed, peers that are wrong *naturally* flip the focal model about twice as often as peers *instructed* to be wrong (29.7% against 14.4%, $p < 10^{-5}$). This survives stratifying on whether the two peers named the same wrong answer, so it is not an artefact of consensus strength. The wrong-answer personas argue in exaggerated faulty styles that are comparatively easy to discount. We now state plainly that our anchored conditions are *conservative* as a probe of corruption risk, and that a real mixed-ability ensemble is likely to be more dangerous than the adversarial one we built — the opposite of the usual concern about adversarial framing, and a direct answer to Tf6M W2 and sTG4 W1.

Reviewer and meta-reviewer preference

We request **the same reviewers and the same Area Chair**. Every point raised is implemented in the manuscript rather than promised, on the identical pool, seed and protocol, and the returning reviewers are best placed to judge whether the prior weaknesses are resolved. The title is unchanged so the paper is recognisable.